

Individual Pathways Activity 10.1

Answers

Transforming and transferring energy

1. Energy is defined as the ability to do work. Energy can be described as either potential or kinetic. Complete the following statements about these forms of energy

- (a) All objects that are moving have kinetic energy.
- (b) All substances and objects possess potential energy. This is also called stored energy.

There are four types of potential energy: chemical, elastic, gravitational and nuclear. Identify the types of potential energy in the examples below.

- (c) A battery has chemical energy.
- (d) The sun generates its energy by nuclear energy.
- (e) When an elastic band is stretched, it has elastic potential energy.
- (f) A pen placed on the end of a bench has gravitational potential energy.

There are five types of kinetic energy: motion, light, sound, heat and electrical. Identify the types of kinetic energy in the examples below.

- | | |
|--|-------------------|
| (g) The energy of runner in the 100 m dash at the Olympics | <u>Motion</u> |
| (h) The energy produced when a trumpet is played | <u>Sound</u> |
| (i) The main energy given off by the blue flame of a Bunsen burner | <u>Heat</u> |
| (j) The energy used to power a toaster | <u>Electrical</u> |
| (k) The main energy given off by a torch | <u>Light</u> |

2. Heat can be transferred by conduction, convection and radiation. In the example below, the sentences have been separated. Rejoin the sentences, and state which type of heat movement the sentence is referring to.

		Conduction, convection or radiation?
When particles very close together, such as in a solid, vibrate ...	... the heat is moved from particle to particle.	<u>Conduction</u>
Heat causes the particles to gain energy ...	... and spread apart, as in a liquid.	<u>Convection</u>
Heat can be transferred without the ...	... need for particles, such as the heat from the sun.	<u>Radiation</u>

3. Using a diagram, describe at least four ways in which a person could make their house more energy efficient.

Answers will vary but may include those listed on pages 317–18 of *Science Quest 8*.

4. In the following matrix, use a tick to indicate whether the objects are luminous or non-luminous. Some objects may be both luminous and non-luminous; for example, when a torch is turned on, it is luminous, but when it is turned off, it is non-luminous.

Object	Luminous	Non-luminous
Sun	✓	
Diamond		✓
Moon		✓
Mars		✓
Light globe	✓	✓
Yellow Bunsen burner flame	✓	
TV screen	✓	✓
Cat's eye		✓

5. How would you make the sound of a guitar string:

- (a) deeper? Use a thicker string that vibrates less, or loosen the string.
(b) have a higher pitch? Tighten the string, or use a thinner string.
(c) vibrate faster? Pluck it faster or harder, or tighten the string.

6. A music teacher was complaining that the music room he was teaching in had too much reverberation.

- (a) What is reverberation?

Reverberation occurs when sound waves are reflected off hard surfaces.

- (b) Explain how the music teacher could overcome this problem.

The teacher could insulate the walls with carpet or felt to reduce the area of hard surface in the room.

7. 'In space, no-one can hear you scream.'

- (a) Is this famous line accurate? Explain your answer.

Yes (although students might raise the possibility that the space suit might cause bias). Theoretically, there are no particles in space to carry sound; therefore, the sound of the scream (ignoring the space suit) would not be carried and so no-one could hear it.

- (b) If the quote was 'In space, no-one can see you scream.', would it still be accurate? Explain your answer.

No. Light travels without the aid of particles and so, if someone were to scream, the action of that scream would be seen but not heard.